

# Visal Chathuranga

AI/ML Engineer · Full-Stack Developer

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## Professional Summary

Aspiring AI Engineer with hands on experience delivering **20+ AI/ML and full-stack projects**, including RAG-based document intelligence platforms, agentic voice AI systems, healthcare EHR solutions, and production grade SaaS applications. B.Sc. in Electrical & Electronic Engineering from the University of Peradeniya *Second Class Upper Division Honours*. Currently developing AI powered healthcare solutions at **MedcubeUSA**, specializing in end to end ML pipelines, LLM integrations, and intelligent automation across FastAPI, Next.js, and cloud-native architectures.

## Technical Skills

|                   |                                                                                                                                                  |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| Languages         | Python ▪ TypeScript ▪ JavaScript ▪ HTML ▪ CSS                                                                                                    |
| Frameworks        | FastAPI ▪ Next.js ▪ Node.js ▪ Flask ▪ Express.js ▪ React ▪ EJS ▪ Electron.js                                                                     |
| AI / ML Libraries | TensorFlow ▪ Keras ▪ PyTorch ▪ Scikit-learn ▪ HuggingFace ▪ LangChain ▪ LangGraph ▪ OpenCV ▪ MediaPipe ▪ Albumentations ▪ timm ▪ LightGBM ▪ NLTK |
| Data & Viz        | Pandas ▪ NumPy ▪ Matplotlib ▪ Seaborn                                                                                                            |
| Web / API Libs    | OpenAI SDK ▪ Prisma ▪ Drizzle ORM ▪ jQuery ▪ Tesseract.js                                                                                        |
| Databases         | MongoDB ▪ PostgreSQL ▪ MySQL ▪ Pinecone ▪ Neo4j ▪ ChromaDB ▪ FAIZZ                                                                               |
| Cloud & DevOps    | AWS (S3 ▪ Textract ▪ Transcribe Medical ▪ EC2) ▪ Vercel ▪ Docker ▪ Kubernetes ▪ Git ▪ CI/CD ▪ DVC ▪ MLflow ▪ RStudio                             |
| Integrations      | OpenAI API ▪ Claude API ▪ Gemini API ▪ Twilio (Voice & WhatsApp) ▪ Stripe ▪ Playwright ▪ JWT ▪ WebSocket ▪ REST APIs                             |

## Experience

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|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| <b>Software Engineer (AI/ML)   MedcubeUSA</b><br>Colombo                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 📅 Sep 2025 – Present  |
| <ul style="list-style-type: none"><li>Architected and deployed enterprise-grade AI solutions including agentic voice assistants, EHR search tools, and intelligent document processing pipelines using RAG, AWS, and Generative AI.</li><li>Delivered end to end full-stack SaaS applications from database schema and REST API design to frontend UI covering healthcare workflows, financial automation, and document intelligence platforms.</li><li>Owned complete product cycles: requirements gathering, system architecture, AI/ML integration, cloud deployment, and post-launch iteration across FastAPI, Next.js, and cloud-native stacks.</li></ul> |                       |
| <b>AI/ML Engineering Intern   GPV Lanka (PVT) Ltd</b><br>Negombo, Sri Lanka                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 📅 Jul 2024 – Oct 2024 |
| <ul style="list-style-type: none"><li>Built AI-powered desktop applications for reflow oven temperature prediction and feeders management using Flask, TensorFlow/Keras, Electron.js, and Python.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                       |
| <b>Automation Engineering Intern   Phoenix Industries Ltd</b><br>Makandura, Sri Lanka                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 📅 Jul 2023 – Oct 2023 |
| <ul style="list-style-type: none"><li>Engineered an automated toothbrush packing machine and a defect detection system for Astra Cups, improving production efficiency and quality control.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                       |

## Projects

### Build Orchestrator: 17-Agent Multi-Agent AI Development System

🔗 Stack: Claude Code ▪ Python ▪ Node.js ▪ YAML ▪ MCP (20 servers) ▪ Bash / PowerShell ▪ MITRE ATT&CK ▪ NIST CSF 2.0 ▪ LangGraph-style Orchestration

- Architected a 17-agent, 7-phase build orchestration system for Claude Code featuring 3 always-on optimization layers token-cost-manager, skill-advisor, and prompt-engineer achieving **30–70% token reduction** per session through intelligent context compression and prompt rewriting.
- Integrated dual skill libraries totalling **2,195+ skills**: 1,441+ Antigravity skills and 754 Cybersecurity skills spanning 26 domains, all dynamically routed via a YAML skill-registry with lazy SKILL.md loading and hash-based session-level caching.
- Deployed a **3-agent security suite** (security-auditor + pen-tester + DFIR analyst) mapped to MITRE ATT&CK, NIST CSF 2.0, MITRE ATLAS, D3FEND, and NIST AI RMF; a polyglot frontend agent supporting 10 frameworks (Next.js priority) and polyglot backend covering Python, Node.js, Go, Rust, and .NET; all backed by 20 MCP servers with per-agent allowlists.
- Shipped 5 slash commands (/agent, /team, /skills, /build-prompt, /init-project), full MITRE-tagged audit trail in docs/runlog.md, graceful degradation across all layers, and cross-platform support (Windows / macOS / Linux) via a self-contained 27-file setup guide.

## VScanMail: AI-Powered Digital Mail Custodian SaaS

**</> Stack:** Next.js ■ TypeScript ■ MySQL ■ Drizzle ORM ■ OpenAI GPT-4o Vision ■ AWS Textract ■ Stripe ■ JWT ■ RBAC

- Built a multi-tenant SaaS platform for scanned mail, cheque processing, delivery requests, and deposits with role-based access control.
- Integrated GPT-4o Vision and AWS Textract for extraction, summaries, tamper detection, signature verification, and smart validation.
- Implemented Stripe subscription billing, 2FA secure auth, audit logs, notifications, and vSendDocs delivery API integration.

[GitHub](#) [Demo](#)

## DocSearch AI: Legal Document Intelligence Platform

**</> Stack:** Node.js ■ TypeScript ■ PostgreSQL ■ MongoDB ■ Pinecone ■ OpenAI Embeddings ■ WebSocket ■ JWT ■ Tesseract.js

- Built an AI-powered legal document search and Q&A platform using a RAG pipeline with OpenAI embeddings and Pinecone, enabling semantic search across case laws, labour codes, and regulations.
- Implemented real-time AI chat with WebSocket streaming, OpenAI Assistants API, and per-session vector stores over uploaded legal PDFs with OCR support.
- Full-stack TypeScript/Node.js app with dual databases, PDF processing, admin/client dashboards, and JWT-secured RBAC.

[Textract Only](#) [+LLM](#)

## Amazon Textract Invoice PDF Processor (Enhanced with LLM)

**</> Stack:** Python ■ AWS Textract ■ AWS S3 ■ Gemini API ■ JSON

- Built a scalable invoice automation pipeline with S3 integration, MD5 duplicate detection, and intelligent table parsing — achieving **95% accuracy** and reducing manual effort by **80%**.
- Enhanced with a Gemini LLM layer to normalize dates, currencies, and payment terms, improving generalization across complex invoice formats.

[GitHub](#)

## MedScribe Pro: Medical Transcription Platform

**</> Stack:** FastAPI ■ AWS Transcribe Medical ■ AWS S3 ■ OpenAI GPT-4 ■ Python

- Built a FastAPI platform converting medical audio (.mp3/.webm) into role-labeled transcripts (Doctor/Patient/Nurse) using AWS Transcribe Medical and GPT-4 for clinical summaries.
- Optimized with S3 Transfer Acceleration, multipart uploads, concurrent workers, and speaker diarization for real-time performance.

[GitHub](#)

## WhatsApp Petty Cash Management System

**</> Stack:** FastAPI ■ Next.js ■ Twilio WhatsApp API ■ AWS Textract ■ OpenAI GPT-4 ■ Docker ■ JWT ■ RBAC

- Built a full-stack expense system with a conversational WhatsApp bot, FastAPI MVC backend, and Next.js admin panel featuring AI-powered OCR and a dynamic rules engine.
- Architected multi-tenant SaaS with RBAC, organization provisioning, audit logging, and dual channels (WhatsApp AI + web) for state-managed claim workflows.
- Deployed Dockerized production system with OCR-validated claims, difference warnings, and a comprehensive admin approval interface.

[GitHub](#) [Demo](#)

## EHR AI Search: Healthcare Intelligence Platform

**</> Stack:** FastAPI ■ React ■ OpenAI GPT-4o Mini ■ WebSocket ■ Vercel ■ Playwright ■ JWT

- Built an AI-powered EHR search platform using GPT-4o Mini with natural language queries, React chat widget, and voice I/O support.
- Deployed as an embeddable Vercel widget with FastAPI backend, WebSocket streaming, and Playwright-based auto-token refresh.

[GitHub](#)

## AI Voice Agent with RAG & Telephony

**</> Stack:** FastAPI ■ OpenAI Realtime API ■ Twilio Voice ■ Pinecone ■ WebSocket ■ Python

- Built a production-ready voice AI system integrating OpenAI Realtime API with Twilio telephony for natural phone conversations with real-time audio streaming.
- Implemented a Pinecone-based RAG pipeline with dual modes: Full Agentic (RAG + web search + tools) and RAG-Only for compliance-critical industries.

## Diabetic Retinopathy Detection: Deep Learning Classifier

**</> Stack:** Python ■ PyTorch ■ EfficientNetB3 ■ timm ■ Albumentations ■ Scikit-learn

- Built an automated 5-class DR severity classifier using EfficientNetB3 with Ben Graham preprocessing, achieving **QWK 0.83** on APTOS 2019 and 0.57 on EyePACS (35K+ images).
- Engineered two-phase training pipeline with mixed precision (fp16), WeightedRandomSampler, patient-level splitting, and Test Time Augmentation.
- Designed a session-resilient training system with auto-resumable checkpointing to Google Drive for free-tier GPU environments (Colab T4/Kaggle P100).

[GitHub](#) [Demo](#)

## YouTube Sentiment Insights

**</> Stack:** Python ■ Flask ■ LightGBM ■ DVC ■ MLflow ■ Docker ■ AWS EC2 ■ GitHub Actions

- Developed an AI-powered Chrome Extension and Flask backend for real-time YouTube comment sentiment analysis, trends, and

dashboards.

- Built a scalable ML pipeline with DVC, MLflow, and LightGBM; Dockerized backend deployed on AWS EC2 with GitHub Actions CI/CD.

[GitHub](#)

## Industrial AI: Reflow Oven Profile Predictor

**</> Stack:** Electron.js ▪ Python ▪ Flask ▪ TensorFlow ▪ Keras

- Designed a hybrid desktop application with Electron.js frontend and Python/Flask backend, integrating TensorFlow/Keras neural networks to optimize industrial reflow oven soldering profiles.

## Education

### B.Sc. Engineering (Hons.) — Electrical & Electronic Engineering

2021 – 2025

GPA: 3.4 / 4.0

University of Peradeniya

Second Class Upper Division Honours

### GCE Advanced Level

2010 – 2018

Dharmasoka College, Ambalangoda

Mathematics: A ▪ Chemistry: A ▪ Physics: B

## Certifications

- Generative AI Mastery with 15+ Real Time Projects — [Udemy](#)
- Deep Learning Neural Networks & Image Processing — [edXcope](#)
- Machine Learning with Python — *with Honors* — [Coursera](#)
- IBM AI Engineering Specialization — [IBM](#) / [Coursera](#)
- Deep Neural Networks with PyTorch — *with Honors* — [IBM](#) / [Coursera](#)
- Building Deep Learning Models with TensorFlow — [IBM](#) / [Coursera](#)
- AI Capstone Project with Deep Learning — [IBM](#) / [Coursera](#)
- Introduction to Deep Learning & Neural Networks — [IBM](#) / [Coursera](#)
- Introduction to Computer Vision & Image Processing — [IBM](#) / [Coursera](#)
- Deep Learning and Reinforcement Learning — [IBM](#) / [Coursera](#)
- NLP with Classification and Vector Spaces — [DeepLearning.AI](#)
- NLP with Probabilistic Models — [DeepLearning.AI](#)
- Introduction to Web Development (HTML/CSS/JS) — [IBM](#) / [Coursera](#)
- Complete Java Full Stack Training — [Digimax Education Sri Lanka](#)

## Extracurricular Activities

|                                                      |                                                     |                     |
|------------------------------------------------------|-----------------------------------------------------|---------------------|
| <b>Treasurer</b>                                     | IEEE Microwave Theory and Techniques Society (MTTS) | Jan 2024 – Aug 2025 |
| <b>Team Member</b>                                   | IEEE MTTS — Programming Team                        | Jan 2023 – Jan 2024 |
| <b>Member</b>                                        | Explorer's Club, University of Peradeniya           | 2022 – Present      |
| <b>Volunteer</b>                                     | Arunella Social Development Initiative              | 2022 – 2025         |
| Supporting school students in underprivileged areas. |                                                     |                     |

## Soft Skills & Interests

**Soft Skills**      Adaptability ▪ Problem-Solving ▪ Teamwork ▪ Communication ▪ Creativity ▪ Time Management

**Interests**      Exploring New Technologies ▪ Playing Chess ▪ Environmental Exploration ▪ Watching Films

## Non-Related Referees

### Dr. Tharindu Weerakoon

Senior Lecturer

Dept. of Electrical & Electronic Engineering

University of Peradeniya, Sri Lanka

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### Dr. Ruwan Ranaweera

Senior Lecturer

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